

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1514

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)*

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Code of Conduct

I M.Yogeshwaran(192524363) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kanish Kumar S

Assessment Tasks

Short Answer Test

SET 1

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.

ANS:

Definition:

Cloud computing is a technology that provides computing services such as servers, storage, databases, networking, software, and analytics over the Internet. Instead of buying and maintaining physical hardware, users can access these resources whenever required and pay only for what they use. Cloud computing offers flexibility, reliability, and cost savings for individuals and organizations.

Characteristics:

1. On-Demand Self-Service:

Users can access cloud resources such as storage, virtual machines, and applications whenever needed without contacting the service provider. This allows quick and easy resource allocation.

2. Broad Network Access:

Cloud services are available over the Internet and can be accessed from laptops, desktops, smartphones, and tablets. This enables users to work from anywhere with an Internet connection.

3. Resource Pooling:

Cloud providers use a shared pool of computing resources to serve multiple customers. Resources are dynamically assigned and reassigned based on user demand, improving efficiency and reducing costs.

4. Rapid Elasticity (Scalability):

Cloud resources can be increased or decreased automatically according to workload requirements. This helps organizations handle sudden increases or decreases in demand without purchasing additional hardware.

2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.

ANS:

- ❖ Cloud computing evolved through several technological advancements over many years. Initially, organizations depended on large mainframe computers that were expensive and shared among many users. As technology improved, personal computers became popular, allowing individuals to perform computing tasks locally.
- ❖ The next major development was **server virtualization**, which enabled multiple virtual machines to run on a single physical server. This improved hardware utilization and reduced infrastructure costs. At the same time, the rapid growth of the Internet made it possible to access applications and data remotely.
- ❖ Later, web-based software and service-oriented architectures allowed users to access applications directly through web browsers. These developments finally led to modern cloud platforms such as **Amazon Web Services (AWS)**, **Microsoft Azure**, and **Google Cloud Platform**, which provide scalable and on-demand computing services across the world.

3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.

ANS:

Server virtualization is one of the most important technologies behind cloud computing. It allows a single physical server to be divided into multiple virtual servers, known as virtual machines (VMs). Each VM operates independently with its own operating system and applications.

Virtualization improves hardware utilization, reduces power consumption, lowers operational costs, and simplifies server management. It also enables quick creation, migration, and backup of virtual machines.

Difference between Virtualization and Cloud Computing

VIRTUALIZATION	CLOUD COMPUTING
Virtualization is a technology that creates virtual versions of computers, servers, storage, or networks.	Cloud computing is a service model that delivers computing resources over the internet.
It allows multiple virtual machines (VMs) to run on a single physical server.	It provides on-demand access to computing resources such as servers, storage, databases, and software
Requires virtualization software (hypervisors) like VMware, Hyper-V, or KVM	Uses virtualization along with automation, networking, and management tools to deliver services.
Main goal is to improve hardware utilization and reduce costs.	Main goal is to provide scalable, flexible, and pay-as-you-go IT services.
Resources are usually managed within a single organization or data center.	Resources are accessed over the internet from cloud providers or private clouds.
Example: Running Windows and Linux on the same physical computer using virtual machines.	Example: Using services from Amazon Web Services, Microsoft Azure, or Google Cloud to host websites or store data.

4. Discuss the role of data centers in delivering cloud services..

ANS:

Data centers are specialized facilities that contain thousands of servers, storage systems, networking devices, and cooling equipment. They are the backbone of cloud computing because they store, process, and manage all cloud resources.

Cloud providers maintain multiple data centers in different geographic locations to ensure high availability and reliability. If one data center fails, another can continue providing services without interruption.

Data centers also provide strong security, regular backups, disaster recovery, load balancing, and continuous monitoring. These features help organizations securely store their data and access cloud services at any time with high performance and minimal downtime.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

ANS:

Infrastructure as a Service (IaaS):

IaaS provides virtual servers, storage, networking, and other infrastructure resources through the Internet. Users manage the operating system and applications, while the provider manages the hardware. Example: Amazon Web Services EC2.

Platform as a Service (PaaS):

PaaS provides a complete development platform where developers can build, test, and deploy applications without managing servers or infrastructure. Example: Google App Engine.

Software as a Service (SaaS):

SaaS delivers ready-to-use software applications through a web browser. Users simply access the software online without installing or maintaining it. Example: Microsoft Microsoft 365 or Google Gmail.

Anything as a Service (XaaS):

XaaS is a broad term that includes any IT service delivered over the cloud, such as AI services, databases, networking, storage, and security. It extends the cloud model to provide almost any technology as a service. Example: Microsoft Azure AI Services.

These service models allow users to choose the level of control and management that best suits their needs while reducing infrastructure costs and improving scalability.

